

ALESSANDRO FILIPPO, PH.D.

STUDENT

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PERSONAL INFORMATION

Born on July 7, 1997 in Rome, Italy.

EDUCATION

Ph.D. (Student, III year)	University of Rome Tor Vergata, Ph.D. Mathematics program. Expected defence date: April-May 2026	From Nov 2022
Master degree	University of Rome Tor Vergata, Pure and Applied Mathematics, 110/110 magna cum laude. Thesis: " <i>Robustness and centrality indices in some complex networks</i> ", under the supervision of Prof. Daniele Bertaccini.	April 2022
Bachelor degree	University of Rome Tor Vergata, Mathematics, 108/110.	Oct 2019

RESEARCH

Interests: Numerical Linear Algebra * Matrix Theory * Network Science.

Research abroad

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| ✈ Aalto University, Espoo, Finland - under the
supervision of Prof. Vanni Noferini, two weeks. | Sept 2025 |
| ✈ University of Strathclyde, Glasgow, UK - under the
supervision of Dr. Francesca Arrigo, three months. | Nov 2023 - June 2025 |

PUBLICATIONS

Published articles

- D. Bertaccini, L. Chiricosta, and A. Filippo. *Source-to-sink communicability: a new centrality measure for directed acyclic networks*. Numerical Algorithms, 2025.
- F. Arrigo, D. Bertaccini, and A. Filippo. *Updating Katz centrality by counting walks*. SIAM Journal on Matrix Analysis and Applications, 2025.
- D. Bertaccini and A. Filippo. *A proposal for ranking through selective computation of centrality measures*. PLOS ONE, 2023.

Submitted / in press articles

- A. Filippo and M. Mazza. *Spectral and computational aspects of a regularized fractional Laplacian for non-local diffusion on graphs*. submitted

Other publications

- D. Bertaccini and A. Filippo. *Notes on Complex Networks*, 2023.
- A. Filippo. *Robustness and centrality indices in some complex networks*, Master thesis, under the supervision of Prof. Daniele Bertaccini, 2022.

TALKS AND POSTERS

- 👤 Mathematics of Data Driven Models, GSSI, L'Aquila, Italy. **Talk:** "On superdiffusive processes on graphs using a regularized fractional Laplacian". Jan 2026
- 👤 DocTorV Junior Seminars, University of Rome Tor Vergata, Italy. Oct 2025
Talk: "Walk -based centrality measures in Complex Networks"
- 👤 NA Seminar, Aalto University, Finland. **Talk:** "Walk-induced strategies for updating Katz centrality and applications to Network Robustness". Sept 2025
- 👤 Modern Perspectives in Approximation Theory: Graphs, Networks, quasi-interpolation and Sampling Theory, Cetraro, Italy. July 2025
Poster: "Updating Katz centrality via walk count".
- 👤 30th Biennial NACONF, University of Strathclyde, Glasgow, UK. **Talk:** June 2025
"A new walk-based centrality measure for directed acyclic networks".
- 👤 2ggALN, University of Pisa, Italy. **Talk:** "Walk-induced Jan 2025
strategies for updating Katz centrality in simple graphs".
- 👤 CACN24, University of Rome Tor Vergata, Italy. **Poster:** "New Dec 2024
numerical strategies for approximately updating Katz centrality".
- 👤 Numerical Linear Algebra, CIRM Luminy, France. **Poster:** "Efficient Sept 2024
strategies for updating some resolvent-based centrality measures".
- 👤 PYSANUM Seminar, University of Pisa, Pisa, Italy. **Talk:** "Complex Dec 2023
Networks: robustness, centrality indices and new strategies for ranking nodes".
- 👤 NA Seminar, University of Strathclyde, Glasgow, UK. **Talk:** "Node Nov 2023
ranking via selective computation of centrality measures".
- 👤 2ggALN, GSSI, L'Aquila, Italy. **Poster:** "A proposal for ranking May 2023
through selective computation of centrality measures".

TEACHING

- 👤 **University tutor at Engineering Department**, Geometry, March 2023 - Jan 2026
University of Rome Tor Vergata, Rome, Italy.
- 👤 **University tutor at Mathematics Department**, Geometry 1, Oct 2021 - Jan 2022
University of Rome Tor Vergata, Rome, Italy.

OTHER SKILLS

- 🌐 **Languages:** Italian (native), English (upper intermediate).
- 💻 **Informatics:** Matlab, Python, C, R, Eiffel.

Last update: 24/02/2026